

Heritage Soundscape and Experiential Sense-of-Place Design in Virtual Reality Towards Heritage Interpretation

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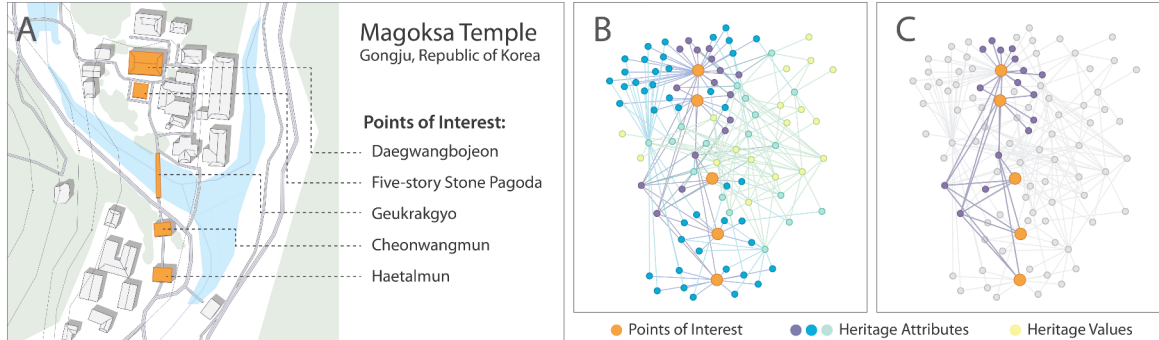


Figure 1: Mapping of the soundscape in Magoksa Temple: (A) Five Points of Interest along the site's main pathway; (B) Network visualisation showing the interconnectivity of all site components; (C) The sound sub-network, highlighting its bridging function between Points of Interest.

ABSTRACT

Virtual Reality (VR) is a powerful tool for heritage interpretation, yet the role of heritage soundscapes remains undervalued compared to visual heritage components. This paper argues for repositioning soundscapes as “interpretive infrastructures” that construct a sense of place in VR. Focusing on Magoksa Temple, a UNESCO World Heritage site in Korea, the study outlines a preliminary design approach: identifying heritage sound elements, articulating their relationships with site components, and translating these into VR design strategies. This direction aims to enhance contextual heritage reconstruction, providing a holistic and immersive user experience in virtual heritage.

Index Terms: Virtual Reality, Heritage Soundscape, Sense of Place, Experience Design, Heritage Interpretation.

1 INTRODUCTION

Virtual Reality (VR) has increasingly attracted the attention of cultural institutions as a communicative and mediating medium for heritage interpretation. Its capacity enables the integrated representation of heritage values, facilitates the transmission of traditional knowledge and historic memory, and improves emotional and spiritual engagement associated with heritage sites within virtual environments (VEs) [6]. These affordances are significant in fostering conservation stewardship, as immersive and affective experiences can strengthen users’ sense of responsibility and long-term commitment to heritage protection [14]. To further enhance these

benefits, current VR studies have moved beyond visually oriented reconstructions that incorporate intangible heritage dimensions to communicate authenticity and integrity as experiential foundations of the sense of place [9]. Heritage soundscapes have been therefore incorporated into VR experiences to improve representations of traditional practices, social relationships, and culturally embedded meanings associated with place [13].

By leveraging sound’s function to bridge the physical gaps among spatially distributed heritage attributes of a site [8, 5], soundscape integration enables a meaningful connection to the past, supporting experiential continuity across temporal and spatial layers of heritage interpretation. Empirical findings in virtual museum contexts, for example, suggest that carefully designed heritage soundscapes and high-fidelity spatial audio are essential for establishing presence, realism, and continuity between past and present [11]. Some studies also demonstrate the role of multisensory and spatially distributed soundscapes in rebuilding place attachment and community connections to sites [3].

However, existing research remains largely focused on the soundscape itself, either by evaluating the auditory interventions’ effectiveness or by addressing technical methodologies related to physical analysis of acoustic parameters, sound reproduction, implementation, and perceived sound quality in VEs [4]. These studies provide valuable insights into acoustic design and technological feasibility—within the ISO definition of soundscape, “the acoustic environment as perceived or experienced and/or understood by a person or people, in context” [7]. Yet, they often overlook the relational nature of heritage soundscapes with a site’s tangible and intangible cultural heritage, thereby limiting their synergetic roles in heritage interpretation [10]. Moreover, user experience evaluations in virtual heritage settings often rely on partial reconstructions, prioritizing emotional response or immersion from a user-centered perspective, rather than situating soundscape experiences within the site’s holistic cultural, historical, and spatial context [11].

To address the above gap, this study presents an exploratory de-

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Table 1: Conceptual Approach to Integrating Heritage Soundscape into Experiential Design.

Dimension	Defining Heritage Soundscape Elements	Identifying Soundscape Linkages to Components of Sites	Recognising the Representational Roles of Heritage Soundscape
Focus	Characteristics	Interpretive Mapping	Virtual Heritage Design
Key Components	- Physical Characteristics: Attributes related to Heritage Soundscape. - Cultural Characteristics: Values explaining the above physical characteristics.	- Heritage Soundscape and Cultural Annotations. - Applying Soundscape to Amplify the Reconstruction of Historical and/or Heritage Scenes.	- Influence of Spatial and Associative Characteristics on Soundscape Design. - Influence of Spatial Functional Features on Soundscape Design.
Final Goal	Enhancing the overall context of a heritage site to communicate its interlinked values and integrity (sense of place) in virtual reality.		

sign direction for integrating heritage soundscapes to support the experiential construction of the sense of place in VR. This design direction will explore the interpretive roles of sound and outline how the identified soundscapes may be incorporated to facilitate culturally and historically grounded sound techniques and their baseline settings in VEs.

2 CONCEPTUAL APPROACH

The term “heritage soundscape” refers to the auditory dimension of a place that embodies collective memory and cultural meaning, shaped through social, cultural, and historical processes [2]. As sound is an integral component of intangible heritage [1], it encompasses sounds perceived and interpreted within historic environments that contribute to place identity, spatial formation, and the cultural image of a site beyond mere physical vibrations transmitted through air. Building on this understanding, this study explores heritage soundscape design in VR as a multi-dimension interpretive process that progressively links acoustic phenomena to heritage meanings and spatial contexts. Accordingly, three inter-related directions—analytically distinct yet operationally concurrent—are provisionally outlined for this study, informed by critical heritage studies, World Heritage studies, VR experience and interaction research (Table 1). The first dimension focuses on defining heritage soundscape elements of a heritage site. The second dimension emphasizes identifying linkages between heritage soundscapes and site components through the development of interpretive mapping. By contextualising soundscapes through cultural annotations and applying them to amplify historical and heritage scenes, this dimension considers how sound mediates relationships between tangible and intangible heritage elements. The third dimension advances toward recognising the representational roles of heritage soundscapes within virtual heritage experience design by examining the influence of spatial, associative, and functional characteristics on soundscape design, positioning sound as a potential design element in virtual heritage interpretation.

3 IMPLEMENTATION

This section reports preliminary insights from a pilot relational mapping conducted at the site level, examining the structural role of sound in heritage interpretation. Section 3.1 addresses the first two dimensions—defining heritage soundscape elements and identifying their linkages to site components—while Section 3.2 advances the third dimension, translating these findings into representational design specifications for VEs.

The selection of the study venue was guided by the identification of a heritage site in which sound is integral to heritage integrity and ongoing cultural practice. Religious heritage sites, which represent a substantial proportion of UNESCO World Heritage List, offer a relevant context: acoustic elements in such settings often structure spiritual experience, mark temporal rhythms of practice, harmonize the sites with the surrounding cultural landscape, and are embedded within ritual observance. On this basis, Magoksa Temple, located in Gongju, Republic of Korea, was selected. The temple is one of

the serial properties of “Sansa, Buddhist Mountain Monasteries,” inscribed in 2018 on the UNESCO World Heritage List [12].

3.1 Articulating the Role of Sound in Heritage Interpretation

Heritage elements and their interconnections were documented through site survey and archival review. Points of Interest (POIs) along the temple’s main pathway—gates (Haetalmun, Cheonwangmun), the Five-story Stone Pagoda, and the Hall of Great Light (Daegwangbojeon)—served as the anchor points for identifying heritage soundscape elements (Figure 1A). The sound dimensions consist of, for example, ritual sounds, natural sounds and environmental acoustics embedded within the cultural landscape.

In our initial mapping, 16 sound-related attributes with one-to-one matching 119 values were revealed from a total of 30 attributes and 312 values of five POIs (Figure 1B), visualized using Gephi. These results, functioning as interconnection of sound’s association across POIs, will be regrouped to create interpretation themes. Sound elements are perceptible across multiple POIs, linking clusters of tangible and intangible heritage (Figure 1C). This bridging characteristic supports the characterization of sound as a communicative mediator: a connector that discloses relational dimensions of heritage value and integrates discrete components into a coherent experiential system. Moreover, Figure 2 demonstrates a preliminary translation of these findings into VR design, depicting the temple courtyard with sound nodes positioned according to their network relationships. This spatialization enables soundscapes to reinforce connections between heritage components, providing the basis for the design outline presented in Section 3.2.



Figure 2: Preliminary spatialization of the relational network within the virtual environment, depicting sound nodes positioned to temple courtyard.

3.2 Design Outline for Virtual Heritage Soundscapes

Following the completion of interpretive mapping, the translation of findings into VE specifications constitutes the current procedural stage of this research. Table 2 presents a preliminary outline

Table 2: Design Outline for Virtual Heritage Soundscapes.

Category	Sub-category	Key Design Strategies
Spatial & Associative Characteristics	Physical Acoustic Geometry	Implementing reverberation and reflection based on spatial scale and materials.
	Symbolic Sound Mapping	Identifying sounds linked to built-heritage and/or historical events/rituals at site components (specific Points of Interest).
Spatial Functional Features	Activity-based Sound	Integrating heritage sound to validate the site's original functions (e.g. religion, rituals).
	Spatio-Temporal Flow	Designing sound transitions (cultural landscape) across different spaces and time-based (diurnal, seasonal & annual rituals) variations.
Communicating Sense of Place	Sound Annotation Linked to Site Components	Synthesizing soundscapes with cultural and historical explanations to provide an additional interpretive dimension.
	Historical Scene Amplification	Additional integration of soundscapes to complement historical scenes, further enriching cultural presence and immersion.

reflecting this mid-stage output, organized according to three categories that address dimensions of soundscape integration for future VE implementations.

The first category, Spatial and Associative Characteristics, addresses physical and symbolic properties of sound. Physical Acoustic Geometry concerns the measurement and rendering of room acoustics based on spatial geometry and the material properties of built structures. Symbolic Sound Mapping involves identification of sounds linked to POIs, encompassing associations with built heritage, historical events, and ritual practices. These specifications translate mapped sound elements into rendering parameters—defining acoustic properties, spatial positioning, and material-based audio calculations for VE reproduction.

The second category, Spatial Functional Features, addresses temporal and activity-based dimensions. Activity-based Sound integrates soundscapes representing original site functions through event-based audio triggers tied to religious observance and ceremonial activity. Spatio-Temporal Flow concerns dynamic audio transitions across spaces and time-based variations, encompassing diurnal cycles, seasonal changes, and annual ritual calendars. These specifications characterize the parameters of soundscape linkages—defining activation triggers, functional weightings, and temporal behaviors—enabling soundscapes to reflect the lived dynamics of heritage sites within the VEs.

The third category, Communicating Sense of Place, addresses interpretive potential within virtual heritage. Sound Annotation synthesizes soundscapes with cultural and historical explanations linked to site components. Historical Scene Amplification integrates soundscapes to complement heritage scenes, enriching cultural presence and experiential continuity. These specifications define the delivery mechanisms through which soundscape meanings are communicated to users, structuring interpretive layers and narratives within the VR experience.

Based on this outline, preliminary VR development integrates identified heritage sound attributes with spatial audio implementation. The three design categories inform distinct technical layers: room acoustic simulation based on architectural geometry, recording POI-triggered audio events linked to ritual and historical associations, and dynamic soundscape rendering responsive to user navigation. To incorporate these technical considerations into context-aware presentation that enhances a sense of space (rather than sound reproduction alone), culturally grounded sound technologies are required. In particular, acoustic metrics that capture the multiple and dynamic interrelations of acoustic events across POIs, along with convincing real-time binaural rendering for six-degrees-of-freedom experiences, require ongoing refinement. This development supports reconstruction of virtual heritage settings for experiential sense of place.

4 FUTURE WORK AND CONCLUSION

This paper presents a preliminary approach to integrating heritage soundscape into virtual heritage design, examining sound's

role as a communicative mediator through relational mapping at Magoksa Temple. Current development proceeds through expert-based refinement of the proposed design outline, with VR reconstruction incorporating 3-dimensional spatial audio rendering. Subsequent phases will extend validation through user studies examining whether sound integration enhances perceived sense of place and site integrity, with potential application to other heritage typologies and investigation of multi-sensory integration. By positioning heritage soundscapes as structural components essential to heritage interpretation, this study challenges visual-centric reconstruction approaches and contributes to comprehensive representations of heritage sites within VEs.

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